

PAPER_I_EXT_AUDIT — 02_orbit_lp_and_reduction

Script 02 — Orbit LP, Symmetrization, Affine Reduction (Gates B & C)

Audit: PAPER_I_EXT_AUDIT · **Script:** 02_orbit_lp_and_reduction.py
Environment: Python 3.14.4, scipy 1.17.1 (HiGHS LP), numpy · seed 20260710

Purpose

Solve, from scratch with an independent LP solver, the linear programs behind the reduction core (§5–7 / ledger L34–L45) and confirm the manuscript's closed forms.

Checks and results

(C0) $q = 0$ branch. The full residual-cover LP gives $M(0) = 0$ for $p \leq 2$ and $p(p-1)/6$ for $p \geq 3$, and $M(0) \geq R(p, 0) - p/2$, for $p = 2 \dots 8$. **VERIFIED.**

(C1) Orbit LP = closed form. For $p = 2 \dots 11$, $s = 2 \dots p$, $q = 1 \dots 12$ (660 pure profiles), the 3-variable orbit LP $\min A\alpha + B\beta + C\gamma$ over the existing triangle constraints equals $\min\{U, D\}$ ($0 \leq 2$) / $\min\{U, D, H\}$ ($0 \geq 3$). **0 mismatches.**

(C2) Orbit symmetrization is lossless. For $p = 2 \dots 8$, the **full** $C(p, 2)$ -variable residual-cover LP (no symmetry assumed) equals the 3-orbit closed form on all 280 pure profiles, $\max |fullLP - formula| = 5.3e-15$. This confirms that averaging an optimum over $Sym(S) \times Sym(K \blacksquare S)$ loses nothing — the crux of Gate C. **0 mismatches.**

(B) Affine reduction / "candidates suffice". For 400 random **mixed** neighborhood profiles on K_p , the full residual-cover LP value $M(\kappa)$ satisfies both $M(\kappa) \geq \min_s M(\kappa^{\wedge}\{S\})$ (concavity reduction, L34) and $M(\kappa) \geq R(p, q) - p/2$ (uniform bound, L55). **0 violations**; minimum slack $M(\kappa) - (R-p/2) = 0.0000$ (the bound is attained — tight — on some mixed profiles, consistent with the extremal family).

Check	Verdict
(C0) $q=0$ branch	VERIFIED
(C1) orbit LP = $\min\{U,D,H\}$	VERIFIED (0/660)
(C2) orbit symmetrization lossless	VERIFIED (0/280, err 5e-15)
(B) affine reduction + uniform bound	VERIFIED (0/400)

Auditor correction (CORR-02-01)

The first run compared the closed form $\min\{U, D, H\}$ against the LP at $q = 0$ and

reported a mismatch at $p=2, q=0$. That was an auditor domain error: the closed form (ledger L45) is the **pure-profile** optimum, and the manuscript (Appendix A.3) states pure profiles have $q > 0$; the $q = 0$ case is the separate §7.5 branch. The test was corrected to run C1/C2 over $q \geq 1$ and to verify the $q = 0$ branch separately (C0). This is not a paper defect — the manuscript is explicit and correct on this point.

Verdict

ALL VERIFIED — no counterexample. The orbit LP closed form, the losslessness of the orbit symmetrization (full LP = 3-orbit LP), and the affine/concavity reduction with its uniform bound $R(p, q) - p/2$ are all confirmed numerically. Combined with Script 01's exact algebra, Gates B, C, E are computationally corroborated (bounded ranges).